

CHIA-PIN WU

EMBEDDED FIRMWARE ENGINEER

PROFESSIONAL SUMMARY

Embedded Firmware Engineer specializing in ARM Cortex-M based industrial control systems and data center liquid cooling solutions. Experienced in Modbus RTU integration, system validation, field troubleshooting, and cross-functional collaboration. Currently expanding expertise in Embedded Linux, Buildroot, BSP, and device driver development.

CONTACT

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LinkedIn: [linkedin.com/in/chia-pin-wu/](https://www.linkedin.com/in/chia-pin-wu/)

GitHub: github.com/LongislandLush

WORK EXPERIENCE

ITG System - Firmware Engineer | Nov 2025-Present

- Developed and maintained STM32G0 firmware for industrial control systems and data center liquid cooling products
- Implemented sensor integration, fan control, valve control, and monitoring functions
- Implemented and maintained Modbus RTU (RS-485) communication architecture, register mapping, and testing
- Performed system bring-up, validation, and field troubleshooting in customer deployment environments

Liyuan Technology - Firmware Engineer | Jul 2025-Sep 2025

- Contributed to handheld printer product development and embedded platform evaluation
- Studied ARM Cortex-M3 (CH32F103) firmware architecture and peripheral control mechanisms
- Explored MIPS and RISC-V based embedded platforms (X2600) and Linux development workflows
- Gained hands-on experience with Linux operation, cross-compilation, and embedded development tools

IoT Training Program Trainee | 2024/11 - 2025/05

- Developed STM32 and ESP32-based embedded applications
- Implemented UART and MQTT communication protocols
- Completed embedded systems and digital design training using Verilog, Quartus II, and KiCad

SKILLS

Embedded Firmware

- C, STM32, ARM Cortex-M, UART, I2C, SPI, PWM, RS-485, Modbus RTU

Embedded Linux

- Linux, Buildroot, Cross Compilation, Makefile

Tools

- STM32CubeIDE, VS Code, Git, Oscilloscope, Logic Analyzer

EDUCATION

M.S., Aeronautics & Astronautics (Control Systems)

National Cheng Kung University | 2012–2015

Master's Thesis:

- "A Study of Diving-Airdrop of Unmanned Aerial Vehicle"

LANGUAGE

- Mandarin: Native
- English: Intermediate (TOEIC 745)
- German: A1 (Goethe-Zertifikat A1)